

Aswanth Bindu Ajith

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LinkedIn

1PROFESSIONAL SUMMARY

Highly motivated interdisciplinary M.Sc. candidate combining a Mechanical Engineering foundation with advanced expertise in data science, machine learning, and scientific computing. Passionate about applying ML models and data-driven decision-making to real industrial manufacturing processes to optimize process parameters, cycle times, and product quality. Backed by practical Agile software development experience and proficiency in Python and PyTorch. Passionate about applying high-performance computing, machine learning and AI to solve complex engineering challenges. Seeking to contribute to Helmholtz-Zentrum Dresden-Rossendorf (HZDR) / CASUS's innovative projects.

2TECHNICAL SKILLS

Python (Advanced),
Scientific Computing,
Numerical Analysis,
C/C++,
Scikit-Learn,
SciPy,
PyTorch,
Pandas

3PROFESSIONAL EXPERIENCE

Working Student Software Developer	<i>Germany</i>
<i>No Border (E-commerce)</i>	Feb 2025 Dec 2025
- Contributed to software development projects within an Agile environment, delivering practical solutions while significantly enhancing Python programming skills - Collaborated effectively within multidisciplinary teams to implement frontend features, demonstrating high adaptability and attention to detail - Supported comprehensive testing and debugging workflows, refining a structured approach to solving challenging technical problems - Utilized Git, GitHub, and Linux for version control and development workflows	

4EDUCATION

M.Sc. High Performance Computing and Quantum Computing	2024Present
Deggendorf Institute of Technology (DIT), Germany	Current Average: 2.6
B.Tech Mechanical Engineering	20192023
APJ Abdul Kalam Technological University, India	CGPA: 6.48

5KEY PROJECTS

Large-Scale Computational Simulation Data Analysis
Developed a Python-based computational simulation to process complex datasets and model interactions among approximately 30,000 system entities. Utilized NumPy and Pandas to extract trends, evaluate data, and identify hidden relationships. Transformed complex numerical data into practical insights and structured scientific reports.
Computing Architecture Optimization Methods
Applied mathematical optimization methods and scientific computing workflows to evaluate computational efficiency of modern architectures. Developed structured analytical approaches to identify computational bottlenecks.
Thermal Analysis of Battery Modules
Conducted extensive engineering simulations to evaluate process behavior and thermal dynamics in high-

dimensional physical systems. Applied mechanical engineering problem-solving to understand impact of operational parameters on product quality.

6RESEARCH INTERESTS

High Performance Computing,
Quantum Computing,
Scientific Machine Learning

7LANGUAGES

English (Professional Working Proficiency),
Malayalam (Native),
German (A1 (Beginner))